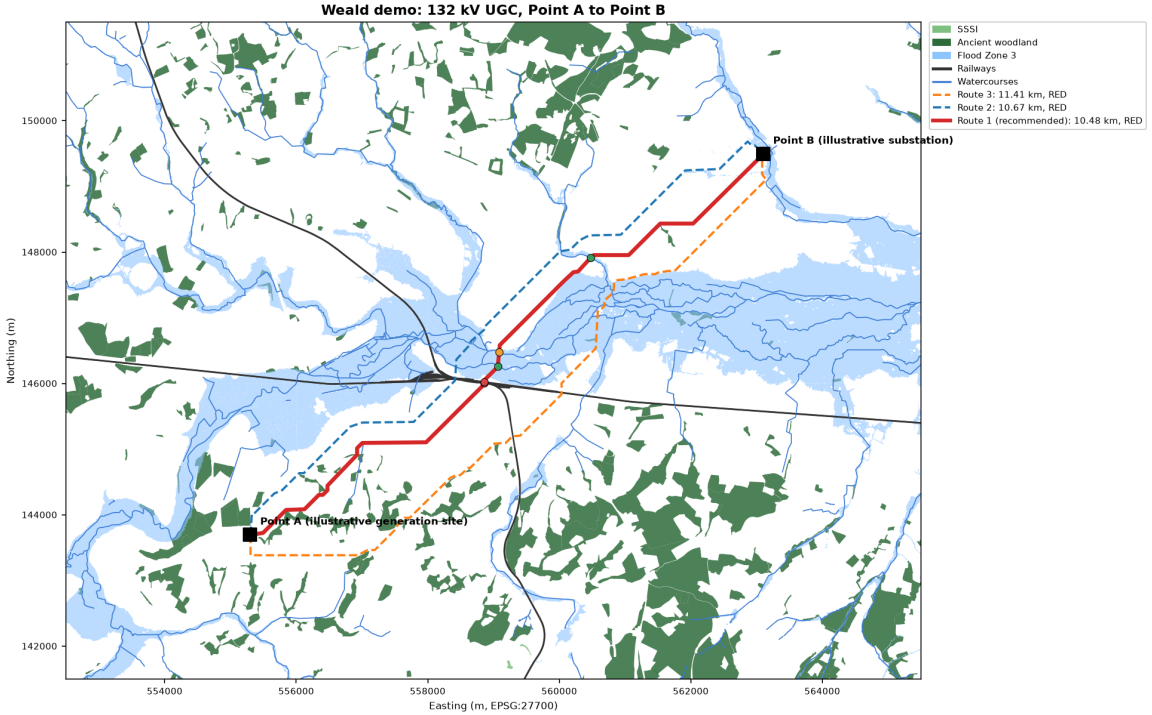
RAG Risk & Crossings Report: Weald demo: 132 kV UGC, Point A to Point B

Illustrative 132 kV underground cable corridor study, Tonbridge area, Kent.
Point A (illustrative generation site) to Point B (illustrative substation). Generated 2026-11-02. Coordinates: British National Grid (EPSG:27700).

Route comparison

Route	Recommended	Length (m)	Weighted cost	Crossings	By type	RED	AMBER	RAG
1	Yes	10481.0	48929.0	5	Railway: 2, Watercourse: 3	3	7	RED
2		10670.0	58281.0	6	Railway: 2, Watercourse: 4	4	13	RED
3		11413.0	62115.0	5	Railway: 2, Watercourse: 3	4	16	RED



Crossings: recommended route (10.48 km)

Ref	Chainage	Type	Class	Name / ref	Angle	RAG	Reason
X01	4+607	Railway	rail	-	89 deg	RED	Railway: rail; parallel features over 8 m (e.g. dual carriageway)
X02	4+625	Railway	rail	South Eastern Main Line	89 deg	RED	Railway: rail
X03	4+950	Watercourse	stream	Gas Works Stream	80 deg	GREEN	Watercourse: stream
X04	5+160	Watercourse	river	River Medway	63 deg	AMBER	Watercourse: river
X05	7+207	Watercourse	stream	Pen Stream	68 deg	GREEN	Watercourse: stream

Constraints, proximity, design and data checks: recommended route

Category	Item	Detail	From	To	RAG
area	Flood Zone 3	464 m of centreline inside	4+732	5+196	AMBER
area	Flood Zone 3	77 m of centreline inside	7+188	7+265	AMBER
area	Flood Zone 3	6 m of centreline inside	10+019	10+028	AMBER
proximity	Within 15 m of ancient woodland (standing-advice buffer)	1 feature(s) within 15 m; closest 14 m	8+539	-	RED
data	Layer 'buildings' unavailable	all Overpass instances failed: https://overpass.kumi.systems/api/interpreter: HTTPSConnectionPool(host='overpass.kumi.systems', port=443): Read timed out. (read timeout=180)	-	-	AMBER
data	Layer 'roads' unavailable	all Overpass instances failed: https://overpass.kumi.systems/api/interpreter: HTTPSConnectionPool(host='overpass.kumi.systems', port=443): Read timed out. (read timeout=180)	-	-	AMBER
data	Layer 'power_lines' unavailable	all Overpass instances failed: https://overpass.kumi.systems/api/interpreter: HTTP 504	-	-	AMBER

Data sources

Layer	Status	Features	Attribution	Fetched (UTC)
sssi	ok	4	Contains Natural England data © Natural England copyright and database right	2026-11-02T13:34:34+00:00
ancient_woodland	ok	378	Contains Natural England data © Natural England copyright and database right	2026-11-02T13:34:34+00:00
flood_zone_3	ok	4176	© Environment Agency copyright and/or database right. All rights reserved.	2026-11-02T13:34:34+00:00
buildings	unavailable	0	© OpenStreetMap contributors, ODbL (production: OS MasterMap Topography)	2026-11-02T13:34:37+00:00
roads	unavailable	0	© OpenStreetMap contributors, ODbL (production: OS MasterMap Highways)	2026-11-02T13:47:06+00:00

Layer	Status	Features	Attribution	Fetched (UTC)
railways	ok	174	© OpenStreetMap contributors, ODbL	2026-11-02T13:58:28+00:00
waterways	ok	310	© OpenStreetMap contributors, ODbL (production: OS / EA main rivers)	2026-11-02T14:03:09+00:00
power_lines	unavailable	0	© OpenStreetMap contributors, ODbL	2026-11-02T14:13:25+00:00

Limitations. Desk-based screening only. Existing buried services (gas, water, telecoms, other electricity assets) are not in open data: a utility search and statutory undertaker enquiries are required. RAG rules are configurable defaults, not a substitute for engineering, environmental or planning judgement.